Lab: 8. Hidden Markov Model

To implement a Hidden Markov Model for weather prediction using the observed umbrella information and calculate the probability of Rain and Sun after each observation.

Given

There are two hidden weather states
- Rain
- Sun

Initial Probabilities: P(Rain) = 0.60 P(Sun) = 0.40

Observations: U: Umbrella N: No Umbrella

Transition Probabilities:

Hidden weather	Tomorrow Rain	Tomorrow Sun
Rain Today	0.70	0.30
Sun Today	0.40	0.60

Observation Probabilities:

Hidden weather	Umbrella	No Umbrella
Rain	0.90	0.10
Sun	0.20	0.80

Step 1: Initial Probabilities

States = ["Rain", "Sun"]

rain = 0.60

sun = 0.40

states store the two possible hidden weather conditions Rain and Sun.

The initial probability of Rain is 0.60

The initial probability of Sun is 0.40

Step 2: Transition Probabilities

rain_to_rain = 0.70

sun_to_rain = 0.40

rain_to_sun = 0.30

sun_to_sun = 0.60

These values show how the weather can change from today to tomorrow

$R \rightarrow R = 0.70$

$S \rightarrow R = 0.40$

$R \rightarrow S = 0.30$

$S \rightarrow S = 0.60$

Step 3: Observation Probabilities

umbrella_if_rain = 0.90

umbrella_if_sun = 0.20

no_umbrella_if_rain = 0.10

no_umbrella_if_sun = 0.80

These values show how likely each observation is for Rain or Sun

$P(U|R) = 0.90$

$P(U|S) = 0.20$

$P(N|R) = 0.10$

$P(N|S) = 0.80$

Step 4: Observations

observations = ["U", "U", "N"]

This list stores the observations for the 3 days in order.

The observations must be processed in order because each day's updated probability is used for the next day's prediction.

Step 5: Process the day at a Time

```
for day in range(len(observations)):
    observation = observations[day]
```

len(observations) gives 3, so range(3) produces 0, 1, 2.
day represents the current day's index.
The loop processes U → U → N one by one

Step 6: Predict the Next Day

```
if day > 0:
    next_rain = rain * rain_to_rain + sun * sun_to_rain
    next_sun = rain * rain_to_sun + sun * sun_to_sun

    rain = next_rain
    sun = next_sun
```

if day > 0 means prediction is done only from Day 2 onward.
Day 1 uses the given initial probabilities 0.60 & 0.40 directly.

Previous belief → Transition Probabilities → Predicted belief

Step 7: Calculate Scores Using the Observation

```
if observation == "U":
    rain_score = rain * umbrella_if_rain
    sun_score = sun * umbrella_if_sun
else:
    rain_score = rain * no_umbrella_if_rain
    sun_score = sun * no_umbrella_if_sun
```

The program first checks whether it is U or N.
If it is U, it uses umbrella probabilities.
If it is N, it uses Non umbrella probabilities.
The multiplication gives a score for each possible weather condition.

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Step 8: Calculate Total Evidence

evidence = rain_score + sun_score

The Rain score and sun score are added together. This gives the total probability of the observed clue.

It becomes the denominator used to convert the scores into proper probabilities.

For Day 1

Rain score = 0.54 Sun Score = 0.08

Evidence = 0.54 + 0.08 = 0.62

Step 9: Normalise the Scores

rain = rain_score / evidence

sun = sun_score / evidence

Each score is divided by the total evidence.

This converts the scores into actual probabilities.

The final Rain & Sun Probabilities add up to 1.

Step 10: Print Result

print ("Day ", day+1, "Observation", observation,
"Rain", round(rain, 4), "Sun", round(sun, 4))

Initial :

Rain = 0.60 Sun = 0.40

Observations

U, U, N

Day 1 : observation = U

day = 0 : 0 > 0 false

⇒ Rain = 0.60 Sun = 0.40

rain_score = rain x umbrella_if_rain

= 0.60 x 0.90 = 0.54

sun_score = Sun * umbrella_if_sun

= 0.40 x 0.20 = 0.08

evidence = 0.54 + 0.08 = 0.62

Normalise: rain = $\frac{0.54}{0.62}$ = 0.8710 sun = $\frac{0.08}{0.62}$ = 0.1290

Day 1 result ⇒ U, rain = 0.8710, sun = 0.1290

Day 2: Observation = U

day = 1 > 0 True

Rain = 0.8710 Sun = 0.1290

next_rain = rain * rain_to_rain + sun * sun_to_rain

= 0.8710 × 0.70 + 0.1290 × 0.40

= 0.6612

next_sun = rain * rain_to_sun + sun * sun_to_sun

= 0.8701 × 0.30 + 0.1290 × 0.60

= 0.3387

rain_score = 0.6612 × 0.90 = 0.5951

sun_score = 0.3387 × 0.20 = 0.0677

evidence = 0.5951 + 0.0677 = 0.6629

Normalise: rain = $\frac{0.5951}{0.6629}$ = 0.8978 sun = $\frac{0.0677}{0.6629}$ = 0.1022

Day 2: result ⇒ U, rain = 0.8978 sun = 0.1022

Day 3: Observation = N

Rain = 0.8978 Sun = 0.1022

next_rain = 0.8978 × 0.70 + 0.1022 × 0.40 = 0.6693

next_sun = 0.8978 × 0.30 + 0.1022 × 0.60 = 0.3306

rain_score = 0.6693 × 0.10 = 0.0669

Sun_score = 0.3306 × 0.80 = 0.2645

evidence = 0.0669 + 0.2645 = 0.3314

Normalise: rain = $\frac{0.0669}{0.3314}$ = 0.2019 sun = $\frac{0.2645}{0.3314}$ = 0.7981